Book CRUD API - Detailed Explanation

1) Imports

FastAPI, HTTPException, Query, status from fastapi; BaseModel, Field from pydantic; List, Optional from typing; uuid4 from uuid

2) App & Schemas

app = FastAPI(title, description, version). Schemas: BookBase shared validated fields; BookCreate inherits; BookUpdate all optional (partial); Book adds id

3) DB

books_db = {} (in-memory dict mapping id -> book)

4) Endpoints

GET / (root), GET /books (query filters: title/author/genre contains, year exact), GET /books/{id} (404), POST /books (uuid, 201), PUT /books/{id} (exclude_unset merge), DELETE /books/{id} (204 or 404)

5) Run & Test

pip install -r requirements.txt; uvicorn main:app --reload -> http://127.0.0.1:8000/docs